# 1 CertificateAuthority

## 1.1 System overview

Native Rust development profile for local Java source `JavaSupportRuntime/`. One independently configured executable, identity, port and state file per role. No Java, Go or Python process implements the service. Some essential operations are explicitly unavailable.

## 1.2 Design and trust boundaries

Tokio/Hyper/Axum with rustls enforces mutual TLS. Certificate chain, validity, local CN/profile and startup-configured optional CRLs are checked. CRLs need process restart to refresh; without CRLs no revocation enforcement is claimed. Tokens are verified with Authentication and bound to the TLS identity. Business calls require ConsumerAuthorization; management requires a verified sysop or explicit broad whitelist. CA provisioning instead applies Java-style named certificate-principal rights. Outbound authorities are pinned to expected CN/profile before credentials are sent, in addition to chain, SAN and time validation.

## 1.3 Persistent state

Independent locked atomic JSON replaces SQL. Clone/validate/fsync/rename commits prevent partially published local transactions. Rename is the commit point; directory-sync failure is an uncertain durable commit, not a rollback. Owner-only keys/password files are separate. Network effects are not distributed transactions; interrupted starts and bridge operations require reconciliation. No automatic Java state import, clustering or deployment is delivered.

## 1.4 Implemented behavior

Native CSR verification/signing, profiles, revocation, PEM CRLs, trusted keys

## 1.5 Open release gates

Explicit SANs only (no CSR extension copying); CRL file publication/reload, OCSP/HSM/HA, full validity/DTO normalization. All systems remain development profiles, not certified interchangeable Java replacements. See `COMPATIBILITY.md` for detailed DTO fields and restrictions; the Java sources remain the authoritative contract.

## 1.6 Produced and consumed services

Advertised implemented HTTPS definitions: `certificateAuthorityManagement`, `certificateSigning`, `certificateValidation`, `echo`, `generalManagement`.

Authorities consumed as applicable: Authentication, ServiceRegistry, ConsumerAuthorization and explicitly configured pinned adapter/provider origins. Only implemented operations are registered. Service versions are 1.0.0 except Gatekeeper/GateTunnel 5.2.2. No AMQP sysMonitor.observe implementation is advertised.

## 1.7 HTTPS operations

JSON unless raw configuration bytes, PEM CRL or echo text. Bodies: 1 MiB; remote responses: 8 MiB. Header: `Bearer IDENTITY-TOKEN//token`. Unknown routes return 404, absent dependencies 503. Unported

operations return 501 after applicable access checks and are excluded from registration. Route presence is not semantic parity.

| Method | Path | Access / status |
|---|---|---|
| GET | /certificate-authority/general/mgmt/get-config | management |
| POST | /certificate-authority/general/mgmt/logs | management |
| GET | /certificate-authority/monitor/echo | certificate |
| POST | /certificate-authority/sign | certificate |
| GET | /certificate-authority/crl | certificate |
| POST | /certificate-authority/checkCertificate | certificate |
| POST | /certificate-authority/checkTrustedKey | certificate |
| GET | /certificate-authority/mgmt/certificates | certificate |
| DELETE | /certificate-authority/mgmt/certificates/{id} | certificate |
| POST | /certificate-authority/mgmt/revocations | certificate |
| GET | /certificate-authority/mgmt/keys | certificate |
| PUT | /certificate-authority/mgmt/keys | certificate |
| DELETE | /certificate-authority/mgmt/keys/{id} | certificate |

Error envelope: errorMessage, errorCode, exceptionType, origin and timestamp. Full Java status, DTO and validation equivalence requires differential testing.

## 2 References and qualification

See `contracts.json` for local Java SHA256 hashes/mappings, `COMPATIBILITY.md` for DTO details and gaps, and `VERIFICATION.md` for tests actually run. No upstream release-tag provenance, complete replacement or real-cloud interoperability is asserted.